

Hyperrealistic VR: A 90-PPD, 1400-Nit, High-Contrast Headset

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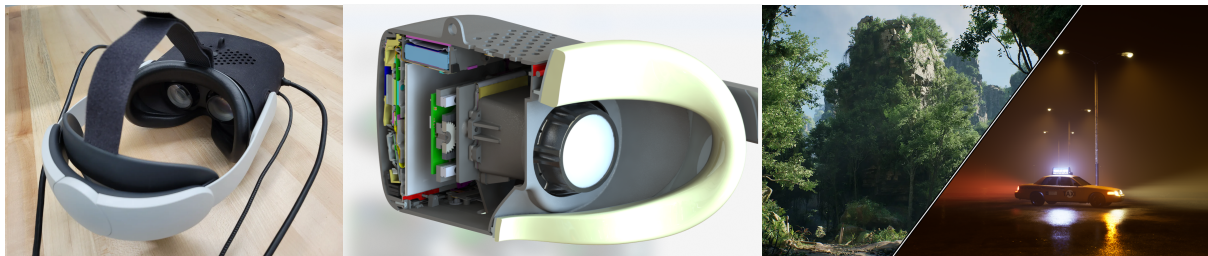


Figure 1: Left: A 90-PPD, 1400-nit, high-contrast VR headset. Middle: This prototype is based on a smooth refractive lens, paired with a uOLED display. Right: Screen captures of high-fidelity rendered scenes, to be shown with this headset.

Abstract

We demonstrate a virtual reality (VR) head-mounted display (HMD) that provides an angular resolution of 90 pixels per degree (PPD), up to 1,400-nit brightness, and high contrast. Leveraging inside-out tracking from the Meta Quest 2, this HMD utilizes an optical design that achieves diffraction-limited performance at 90 PPD. By utilizing Unreal Engine 5 and Nvidia's DLSS 3 AI supersampling method,

hyperrealistic demos can be supported at interactive frame rates. We present this Emerging Technologies installation to exhibit the combined visual benefits of ultra-high-resolution, high-brightness and high-contrast displays. To our knowledge, this work documents the first VR HMD that achieves this combination of visual benefits in a manner supporting a natural sense of visual immersion — setting a new milestone in terms of how realistic VR experiences can be with state-of-the-art technologies.

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1 Introduction

What is keeping us from having a VR display that is almost indistinguishable from being in the place physically? This has been our research focus, and has been framed as “What would it take to pass the visual Turing test?” [Lanman 2023]. For a VR headset, jointly advancing both the display hardware and graphics software is vital to achieving this goal.

Four aspects limit VR display hardware from faithfully projecting realistic images: resolution, high dynamic range (HDR), optical distortion, and natural accommodation (depth of focus). Matsuda et al. previously demonstrated the impact of high brightness [Matsuda et al. 2022], and Zhao et al. previously showcased retinal resolution with accommodation support [Zhao et al. 2023]. This demo focuses on combining higher-than-retinal resolution with HDR displays and minimized optical distortions. To fully exhibit the visual benefits of these hardware advances, we pair our prototype headset with high-performance graphics for visualizing highly detailed assets, allowing photorealistic imagery to be driven interactively.

2 Design Choices for Hyperrealistic VR

Figure 2 reviews key display parameters that impact realism. Compared to Quest 3, our prototype features over 14× brightness (up to 1,400 nits vs. 100 nits) and 3.6× resolution (90 PPD vs. 25 PPD). To achieve the highest VR resolution we are aware of, field of view (FOV) is reduced to $33^\circ \times 33^\circ$, which is considerably narrower than today’s consumer headsets. However, we find that this FOV is still sufficient to achieve a step change in visual realism, preserving binocular vision in the central FOV. Image aberrations and pupil swim are minimized over this FOV, and, due to the use of coated refractive viewing optics and uOLED displays, our headset has much higher contrast than Quest 3.

Why do we need more than 60 PPD for this headset? A target “retinal resolution” of about 60 PPD has long been held as the gold standard for direct-view displays — such as monitors and mobile phones — being sufficient to depict the 20/20 line on an eye chart. When rendering text on such displays, the edges of characters can be deliberately aligned to minimize splitting pixels. However, VR scenes are generated in a world-locked environment, and text can no longer regularly align with display pixels. In VR, 120 PPD or higher may be needed to support 20/20 vision. We note that, with our 90-PPD prototype, many individuals report that they can perceive a difference in resolution, in comparison to earlier 56-PPD prototypes [Zhao et al. 2023].

3 Display System, Headset and Software

Our viewing optics consist of three singlets made of high refractive index crown/flint glass to correct monochromatic and chromatic aberrations. The design nearly achieves diffraction-limited performance, providing a contrast above 0.5 at the Nyquist frequency (45 cycles per degree) across the full FOV. All lens elements are custom made with anti-reflection coatings to enhance contrast and maximize brightness. To provide world-locked content, the headset

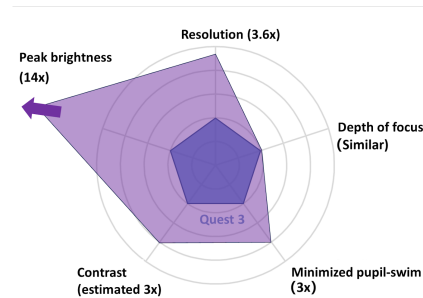


Figure 2: Comparisons of key display hardware capabilities impacting realism, depicting those of Quest 3 (blue) and our hyperrealistic prototype (purple).

is tracked with six degrees of freedom using inside-out tracking provided by hardware from Meta Quest 2.

The 90-PPD resolution of our prototype offers a unique challenge in rendering hyperrealistic environments. It requires incredibly highly detailed assets, creating significant VRAM and rendering performance challenges. For our exhibition, Unreal Engine 5 (UE5) is used to achieve photorealistic models and lighting, leveraging photogrammetry from Quixel Megascans. These photorealistic assets utilize high-resolution source textures and geometry. UE5’s Nanite converts the high-complexity geometry into a performant version with no notable impact on visual quality, and is further combined with advanced virtual texturing to enhance performance. UE5’s high-fidelity lighting solution, Lumen, uses path-tracing and ray-tracing to achieve realism, but at very high performance cost. Nvidia DLSS 3, AI supersampling method is applied to mitigate cost and temporal artifacts of Lumen, in addition to upscaling the image. The headset’s high resolution further empowers the DLSS algorithm, as it generally performs better at higher resolutions.

4 Demo Experience

Users will start from a Quest 3 headset as the baseline, which renders the same high-fidelity scenes, before experiencing these assets with our hyperrealistic VR prototype. Users are placed into different scenes, including a virtual nature environment where they can examine natural objects and trail maps to experience the high level of visual realism and fine text legibility, respectively.

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